



Fig. 1: IC design flow chart

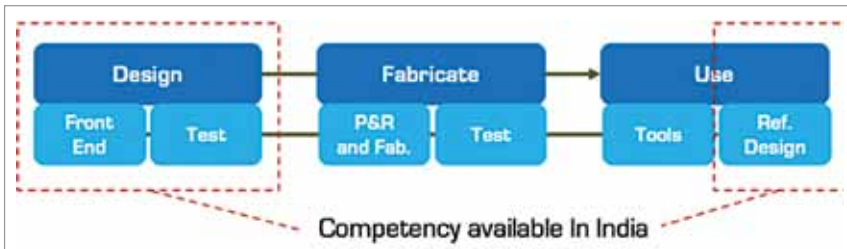


Fig. 2: Boxes encircled in red show what can be done in India

known as foundries—to manufacture the ICs. Fabrication process has four steps before it is converted to individual ICs:

(i) First process in fabrication is the Place & Route (P&R) (also known as back-end), which is the equivalent process of PCB routing. P&R process is very specific to each foundry as it is dependent on semiconductor process used by the foundry.

(ii) P&R output is converted to masks and the ICs are fabricated in a large 20.3cm or 30.5cm (8-inch or 12-inch) wafer. A wafer will have multiple ICs known as dies.

(iii) Foundries test each of the ICs (dies) and mark the good and the bad ones in the wafers.

(iv) Typically, foundries deliver the tested wafers to the OSAT (outsourced semiconductor assembly & testing) vendors. These OSAT vendors cut the individual ICs, package them based on the chip vendors' design, test and mark them and deliver. OSAT is typically an intensive manual process, and most semiconductor foundries outsource this to companies who specialise in this service. Malaysia, Vietnam, and Indonesia are the major OSAT locations.

3. At this stage the IC is ready and can be used for designs. Life would be easy if the IC happens to

be a simple device like an op-amp. (This is not true as you still need a datasheet with specifications to use the device). But if the IC happens to be a microprocessor, the life is different. Any microprocessor that is introduced in the market needs three critical items as a must to succeed:

(i) Foremost are the software development tools consisting of assemblers, compilers, integrated development environment (IDE) to help the software developers to work with the processor. Tools have to be efficient in terms of compilation efficiency, optimal (size) compiled code, and libraries for the peripherals in the devices.

(ii) Second part of the tools is the operating system support. Typically, a standard OS like Linux and real-time operating systems like FreeRTOS should be ported on to the processor and delivered as ready-to-use component for the designers to focus only on the application development. Along with this board support package (BSP), a piece of firmware which allows the OS to work without worrying about the underlying processor variation, should be ported.

(iii) With focus on reduced time-to-market, another support item that is expected by the designers is the reference designs that

demonstrate the hardware features of the processor. These reference designs have to be a total package covering the schematics files, PCB CAD files, Gerber files, and the firmware code that runs on the reference board. This helps the designers understand the design and develop their designs. Many times, designers use this for their proof-of-concept prototypes.

Now we can see what processes can be done in India to develop a processor from scratch. Considering the current situation in India, Fig. 2 shows what can be done in India and what can't be! The ones under the red boxes are skills that are available in India. For the rest we need infrastructure.

We can see that while we have good competency in designing, we have very less in the subsequent stages—barring some experience in the reference design. Having understood the current status let us see why we need an Indian processor.

## Need for Indian processor

This was the main reason for this article. Three key elements that drive the need for an Indian processor are:

1. Primary reason is the security. With the proliferation of connected devices ensuring end-to-end security is the most critical issue. This calls for secure communication (encryption) and authorised devices. While the encryption is easy to implement, designing authorised/certified devices calls for certifying every device and the authentication mechanism must be located inside India. This ecosystem is extensive and needs to be installed before the production of ICs starts. Today, even though authorised/certified devices (also called trusted devices) are available from global vendors, the device data is located outside India, which is a security risk.

2. Avoiding supply chain risks. As we have seen, severe shortage